A classic Seyfert 1 galaxy started showing strong **quasi-periodic** flux variations…

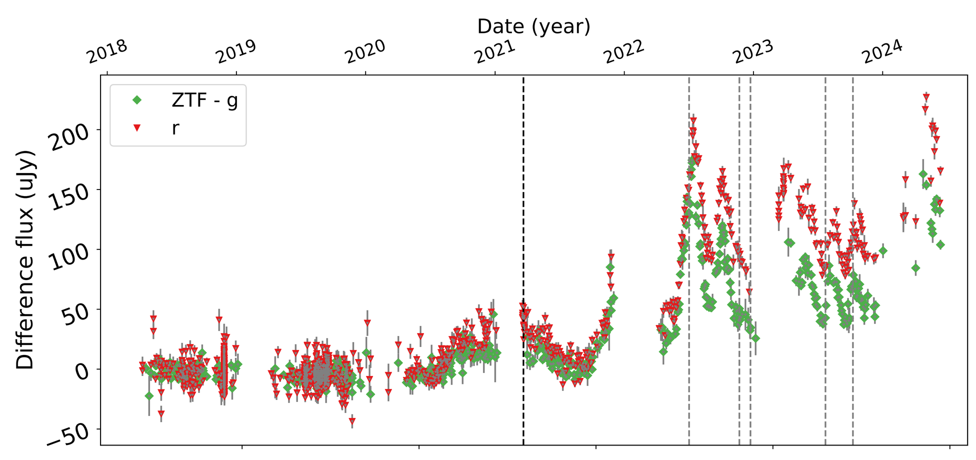
Appears to be a **sub-milliparsec** massive black hole **binary** accreting a gas cloud!

AT 2021hdr: A candidate tidal disruption of a gas cloud by a binary super massive black hole system

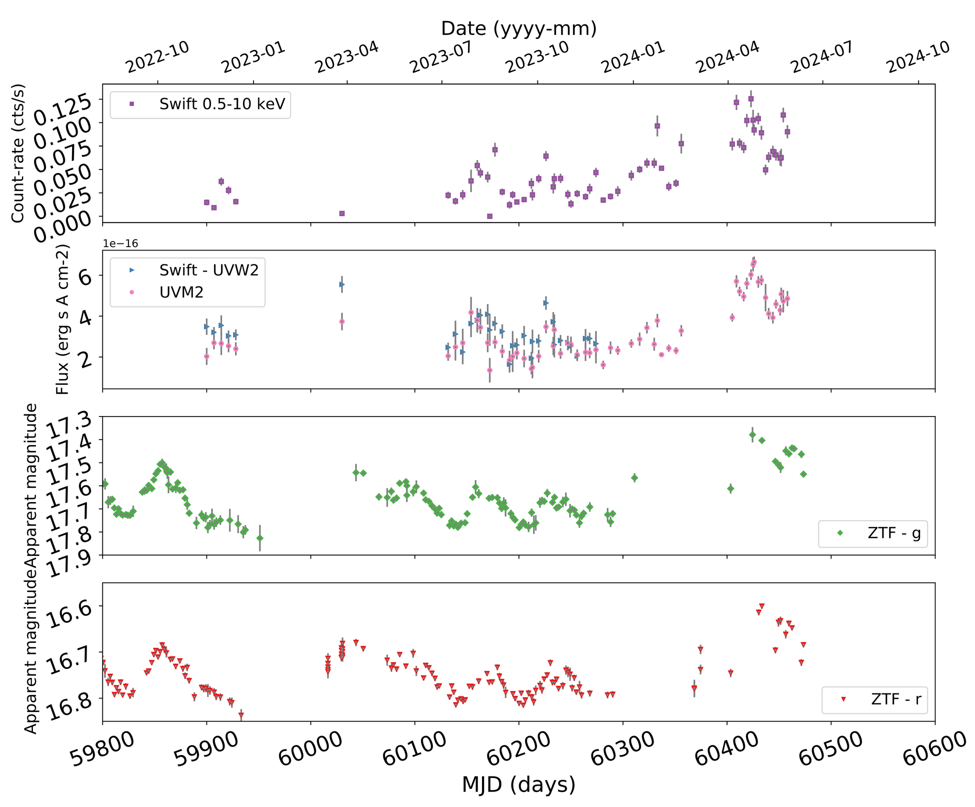
L Hernández-García, AM Muñoz-Arancibia, P Lira, G Bruni, **J Cuadra**, P Arévalo, P Sánchez-Sáez et al.
A&A, 692, A84 (2024)

Discovery and follow-up

AT 2021hdr was a Seyfert 1 galaxy with no unusual features. It had standard stochastic, low-level variability. In 2021 it started to show larger, 0.2-mag oscillations which seem periodic.



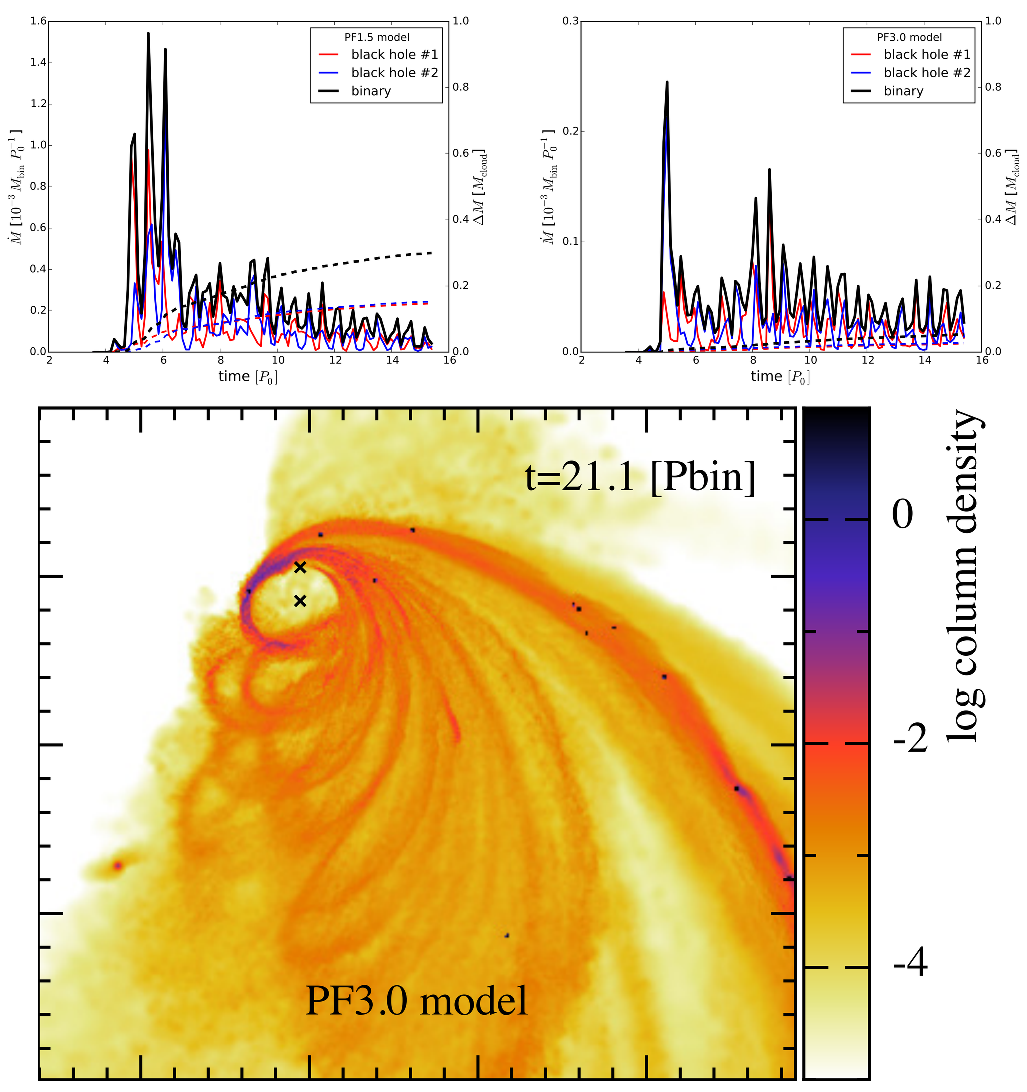
Similar oscillations observed in X-rays and UV.



Nature of the source

Observed data defy standard categories. X-rays are too hard compared to QPEs and TDEs. Optical spectrum remains constant, so no "changing-look" AGN. Temporal behaviour doesn't match either. SMBH binary can't easily explain sudden start.

Earlier simulations of a binary accreting a gas cloud (Goicovic, Cuadra et al 2016, shown below) do match qualitatively the observed light curve features. This scenario is also consistent with change in colour due to reddening by cloud debris.



Binary interpretation

From the oscillations we assume an orbital period of $\sim$ 130 days. The optical spectrum requires a total black hole mass $\sim 4 \times 10^7\,M_\odot$. Together they imply a separation of $\sim$ 0.83 mpc $\sim$ 220 – 430 $R_{\rm Sch}$ $\sim 0.5\mu$as – too small to resolve :-(

Time-scale for merger is just $\sim 7 \times 10^4$ yr, making this source very relevant for the study of hierarchical growth of SMBHs in the Universe.

Ask me for more details!

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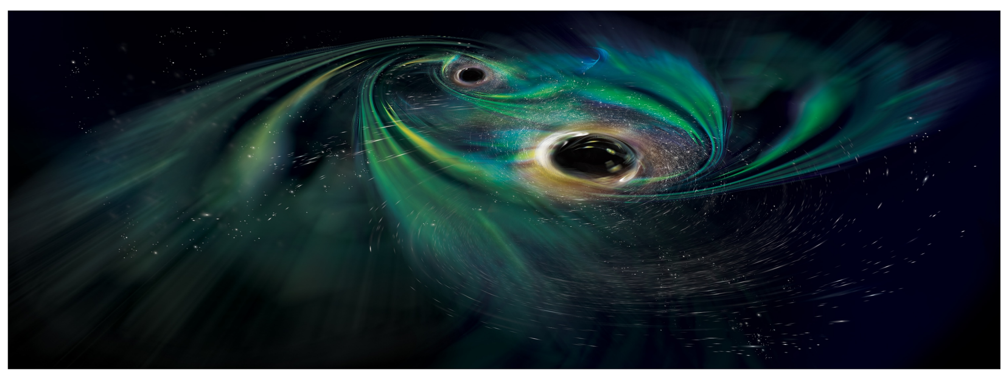

JC acknowledges financial support from ANID – FONDECYT Regular 1251444, and Millennium Science Initiative Program NCN2023_002.

Scan QR code to get the paper and watch the simulations!

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